1 Problem finding

The Think shelf names three sources and all three are about solving rather than finding. Pólya gives a procedure for working a problem once you have one. Poincaré describes the two faculties a mathematician works with. Tao describes when intuition becomes trustworthy. None of the three tells you which problem to take up.

That gap is the subject here, and the instrument for it already exists in this project. The Headspace page is a problem-finding artifact, and its structure is worth naming so it can be used deliberately.

1.1 Why a list of heuristics is not the deliverable

The Mental World Modeling thread states the obstacle in its own words, and it applies to problem finding as much as to problem solving.

"General heuristics ('work backwards,' 'try a simpler case') transfer badly: people who can recite them still fail to deploy them, because knowing an operation and regulating when to use it are different capacities."

The Next page names the same thing as the under-modeled part: "The evidence is consistent that people fail less for lack of heuristics than for lack of regulation, choosing the next move, noticing a line isn't working, deciding when to abandon it. That control layer is a distinct problem from the operations it governs, and it is the under-modeled one."

So this section gives a procedure with stopping conditions rather than a list of moves. The stopping conditions are the part that carries the work.

1.2 Poincaré: which faculty is being used

Poincaré separates two kinds of mind and assigns each a job. The distinction is useful here because problem finding and problem solving draw on different ones.

"Thus logic and intuition have each their necessary role. Each is indispensable. Logic . . . is the instrument of demonstration; intuition is the instrument of invention."

The two work differently. Logical minds advance "step by step, after the manner of a Vauban who pushes on his trenches." Intuitive minds make "quick but sometimes precarious conquests, like bold cavalrymen of the advance guard."

Each has a failure. Analysis "can have, therefore, no tool other than the scalpel and the microscope," yielding "a multitude of procedures" without showing "which will lead us most promptly to our goal." Intuition alone "can not give us rigour, nor even certainty."

The passage that matters most for reading and for reviewing: verification is not understanding. "When we have examined these operations one after the other and ascertained that each is correct, are we to think we have grasped the real meaning of the demonstration? Shall we have understood it . . . ? Evidently not."

Problem finding is invention, so it runs on the faculty Poincaré says cannot give certainty. That is not a reason to distrust it. It is a reason to write the candidate down and test it rather than acting on it directly.

1.3 Tao: when your intuition is worth acting on

Tao's three stages answer the question Poincaré leaves open, which is when an intuition has earned the right to direct work.

The pre-rigorous stage is one "in which mathematics is taught in an informal, intuitive manner, based on examples, fuzzy notions, and hand-waving," with emphasis on computation rather than theory.

The rigorous stage is one "in which one is taught that in order to do maths 'properly', one needs to work and think in a much more precise and formal manner," where "one is expected to be able to comfortably manipulate abstract mathematical objects without focusing too much on what such objects actually 'mean'."

The post-rigorous stage is one "in which one has grown comfortable with all the rigorous foundations of one's chosen field, and is now ready to revisit and refine one's pre-rigorous intuition on the subject, but this time with the intuition solidly buttressed by rigorous theory."

The sentence that decides what rigor is for: "The point of rigour is not to destroy all intuition; instead, it should be used to destroy bad intuition while clarifying and elevating good intuition."

The Teach page already states the failure this framework names, and states it as a design commitment: "Stalling at the second stage, with intuition discarded as unrigorous, is the failure I design against."

The consequence for problem finding. An intuition about which problem matters is worth acting on in proportion to how much formalism has already passed through it. An intuition that has survived formalization is one that can be built on. An intuition that has never met the formalism is a candidate, not a direction.

1.4 Pólya, and where his procedure starts

Pólya's four phases are already in use in this project as a reading lens, applied across seventeen sources in `reading_insights.md`: understand the problem, devise a plan, carry it out, look back.

Two things about that use are worth extracting.

The phases are applied to *other people's* papers, to see how each one decomposed its problem. That turns Pólya from a solving procedure into a reading procedure, and the reading is where the candidate problems come from.

The heuristic that does the most work in those notes is the one about recognition. Applied to prior cyber environments, it surfaces as a critique: previous work is faulted for unrealistic goals, engineered rewards, and leaking the action space. The note records the move as restating by contrast, and it is the operation that converts reading into a problem.

Pólya's procedure begins after a problem is in hand. The look back phase is the one that produces new problems, and in this project it is instantiated as turning each design choice into a direct answer to a named prior failing.

1.5 The Headspace page, read as an instrument

The Headspace page has a fixed shape and the shape is the method. Each entry is a question in the author's own words, with sources listed under it. Some questions carry sub-questions that narrow the parent.

Four properties make it work.

The question is written before the answer exists. Entries such as when is a cybersecurity decision a reinforcement learning problem, or what is mind and what is muscle, have no answer under them. They have sources that are relevant to them.

Sources are attached to questions rather than to topics. A paper appears under the question it speaks to. The same paper could sit under two questions, and where it does, that is a finding about the questions.

Questions are stated at a level where they could be wrong. When does an agent have the need to model the world in the first place is answerable. What is interesting about world models is not.

Sub-questions narrow rather than elaborate. Under the opponent modeling entry, the sub-questions ask what defenders want to predict about adversaries, what the reality of data availability is, whether zero-day defense is a zero-shot or a few-shot challenge, and whether existing opponent modeling can differentiate human from AI adversaries. Each one could be taken up as a project.

1.6 A procedure for locating a topic

Question 32 in the open questions asks to triangulate the pinpoints of cyber attacker modeling against opponent modeling advances and challenges, and the author's own note on it asks for a pseudocode of that operation. This is that procedure.

Inference. This is assembled from the sources above and from the Headspace structure. No source states it.

1. **Write the question so it could be wrong.** If no observation would settle it, it is a topic and not a question. Rewrite until it names something that could turn out either way.
2. **Name the two fields it sits between.** A question worth taking up usually sits where two literatures make different assumptions about the same object. Attacker modeling and opponent modeling are one such pair.
3. **Find what each field takes for granted that the other does not.** This is the census operation from the lexicon method, run across two fields rather than across one project. Where the two use one term in two senses, that is a candidate.
4. **Check the demand side.** Ask what a practitioner in the applied field wants predicted, and whether the data to predict it exists. The Headspace sub-questions do this explicitly: what do defenders want to predict about adversaries, and what is the reality of data availability.
5. **Check that the answer would change something.** From the wording checklist: a number earns its place by changing a decision. A question earns its place the same way. Name what would be done differently under each answer.

6. **Check whether it is reachable.** A question is reachable when an experiment you could run this year would move it. Otherwise it is a five year question and belongs on the vista rather than in an agenda.

7. **Stop when a round of reading adds no new candidate.** Same stopping rule as the existence search. If a complete pass over the two literatures produces nothing new, the space has been mapped, and what remains is choosing rather than searching.

The four tests in steps one, five, six, and seven are the regulation layer. They are what the Mental World Modeling thread says people lack, and they are the reason this is written as tests rather than as prompts.

1.7 Sources

- Henri Poincaré, *Intuition and Logic in Mathematics*, read at `mathshistory.st-andrews.ac.uk/Extras/Poincar` Quotations are exact as returned by that page.

- Terence Tao, *There's more to mathematics than rigour and proofs.* **Primary not retrieved.** The site returned 403 on two attempts and the archive was unreachable. The three stage quotations here come from a search summary of that page and are marked secondary. They should be checked against the original before being quoted in anything public. The author's own restatement of the same framework is on the Teach page and is quoted from there directly.

- George Pólya, *How to Solve It.* Not opened for this file. The four phases and the recognition heuristic reach this document through their use as a reading lens in `Literature/reading_insights.md`, question 2, applied across seventeen sources.

- The Mental World Modeling overview and its Next page, and the Headspace page, all read in full from the site.

The procedure in the previous subsection is an extension. It applies the Headspace structure and the lexicon census to the problem of choosing a topic, which none of the named sources does.